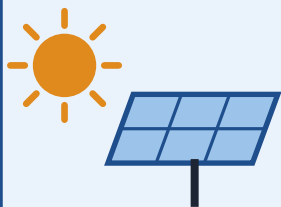


Physical PV data



Yulara 768 kW
off-grid, desert

NIST 271 kW
grid-connected, humid

Curtailed series removed
≈50% of its energy curtailed



Archived GFS forecasts
only runs published in time

Honest selection (AHSE)

LightGBM

XGBoost

PatchTST

N-HiTS

LSTM

GRU

+ GFS inputs for tree models

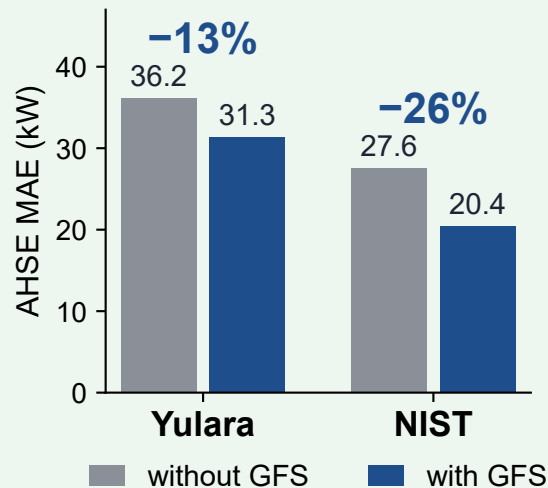
per lead-time block: 1 h, 1–6 h, 6–24 h

**combine models only if the
bootstrap gain is > 0**

otherwise: best single model

Decided on validation data only
+ out-of-fold conformal 90% intervals

Test results, 1–24 h ahead



90% intervals: coverage 0.90–0.92

AHSE never significantly worse than best model